THARUN DUGGI

AI Agent Developer | Python Full Stack Engineer | RAG & LLM Workflows | AIDLC

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PROFESSIONAL SUMMARY

Results-driven AI Agent Developer and Python Full Stack Engineer with 3+ years of experience architecting production-grade Generative AI applications, multi-agent orchestrations, and high-performance backend systems. Expert in building low-latency Retrieval-Augmented Generation (RAG) pipelines, autonomous AI workflows, and scalable enterprise platforms using LangChain, FastAPI, and Flask. Proven track record in infrastructure cost optimization, successfully migrating production workloads from legacy AWS EC2 setups to serverless AWS Lambda architecture—cutting hosting spend by over 55% while maintaining 99.9% uptime.

EXPERIENCE

JAN 2026 - PRESENT — **PYTHON AI AGENT DEVELOPER**

Techgy Innovations Pvt Ltd

- Engineered production-grade RAG pipelines using LangChain and advanced Vector indexing, enhancing context-retrieval accuracy and reducing LLM hallucinations across autonomous AI agents.
- Optimized cloud infrastructure architecture by identifying excessive operational spend on heavy AWS EC2 instances and executing a strategic migration to serverless AWS Lambda, cutting production hosting costs by over 55%.
- Architected scalable backend systems and high-throughput RESTful APIs using FastAPI and Flask, improving end-to-end data transmission efficiency by 25%.
- Spearheaded intelligent agentic workflows and secure webhook integrations, automating 40% of complex, multi-turn user qualification steps and manual data logging.
- Refactored MongoDB and MySQL databases for low-latency streaming, accelerating real-time conversational response generation across active client platforms.

AI/ML & Agentic Alignment Highlights:

- Applied AIDLC from requirements and solution design through development, validation, and deployment.
- Used Claude-based memory & skillsets across the agent lifecycle for iterative development and refinement.

JAN 2023 - JAN 2026 — **PYTHON DEVELOPER**

Innoclique Cognitive Health Technologies

- Developed enterprise-grade backend applications using Python and Flask for mission-critical healthcare management systems serving over 10,000 active users.
- Integrated specialized LLM solutions via LangChain, introducing contextual history tracking that slashed customer support response latency by 60%.
- Built robust web automation frameworks applying Object-Oriented Programming (OOP) and Selenium, reducing manual business process workloads by 75%.
- Deployed localized RPA desktop scripts using PyAutoGUI to seamlessly synchronize 1,000+ daily financial transactions with core relational databases.
- Optimized complex SQL query performance and schemas by 40% through precise index tuning and architectural normalization.

AI/ML & Automation Highlights:

- Applied AIDLC from requirements and solution design through development, validation, and deployment.
- Used Claude-based memory & skillsets across iterative automation development and refinement.

KEY PROJECT CONTRIBUTIONS

AI Agent Platform Development (Serverless Multi-Tenant RAG Pipeline)
Applied AIDLC across requirements, design, development, validation and deployment, using Claude-based memory & skillsets.
Deployed a multi-tenant AI agent platform built on serverless principles to handle 500+ daily conversations seamlessly. Integrated WhatsApp Business API with intelligent RAG routing to automate lead qualification and deliver real-time product answers. Implemented an automated conversational summarization pipeline extracting structured metrics from unstructured client interactions.
Impact: Reduced response overhead and automated end-to-end customer onboarding and lead qualification.
Tech Stack: Python, LangChain, FastAPI, LLM, WhatsApp Business API, Vector DB, MySQL, AWS Lambda

Healthcare Claims Automation Engine (Page Object Model Suite)
Applied AIDLC across requirements, design, development, validation and deployment, using Claude-based memory & skillsets throughout.
Formulated an end-to-end web automation suite using the Page Object Model (POM) in Selenium to speed up claim handling workflows by 75%. Established robust data verification rules to identify and filter broken claims before submission.
Impact: Accelerated claim handling throughput while maintaining zero submission error rates.
Tech Stack: Python, Selenium WebDriver, Page Object Model (POM), MySQL, OOP

SKILLS

Languages: Python, JavaScript, SQL (Advanced), HTML/CSS

AI/ML & GenAI: AI Agent Frameworks, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Agentic AI Workflows, LangChain, Prompt Engineering, Vector Databases, Context-Aware Agents

Frameworks & Tools: FastAPI, Flask, Selenium WebDriver, Playwright, PyAutoGUI, RESTful APIs

Cloud & Production: AWS Lambda (Serverless Migration), Amazon EC2, Amazon S3, Docker, Git, CI/CD Pipelines, Webhooks, Microservices Architecture

Databases: MySQL, MongoDB, Vector Indexing / Storage

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering (Aug 2018 - Aug 2022) — Hindustan University
Professional Certifications: Developing LLM Applications with LangChain (DeepLearning.AI) | Advanced Python Programming & OOP | Flask & FastAPI Architecture | Automated Testing with Selenium WebDriver